1 Question 1

For very small values of w (e.g. $w = 1$, $w = 2$), the embeddings are based only on close neighbors and thus reflect a very local structure of the graph. These embeddings can capture finer details about nodes and their local/direct neighborhood, but are also more sensitive to noise. Moreover, they may miss broader information and global context of the graph.
In contrast, for very large values of w (e.g. $w = 14$, $w = 15$), the embeddings incorporate a much larger range of nodes and therefore reflect a broader, global structure of the graph. These embeddings capture nodes that are farther apart in the graph and thus encode longer range relationships. However, since nodes that are far from each other may be blended into the same context (embedding), the model can struggle to distinguish local structures, which can lead to a loss of local pattern details.
In their experiments on node classification, [1] show that moderate window sizes (e.g. $w \in \{6, \dots, 9\}$) achieve the highest classification accuracies. These results confirm that the embeddings should balance between both local and global patterns of the graph.

2 Question 2

We observe the two followings:

$$X_1 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 1 \\ -0.5 & 0.5 \\ 0.5 & -0.5 \\ 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \qquad X_1 \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 1 \\ -0.5 & 0.5 \\ 0.5 & -0.5 \\ 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \cdot 1 + 1 \cdot 0 & -1 \cdot 0 + 1 \cdot (-1) \\ -1 \cdot 1 + 1 \cdot 0 & -1 \cdot 0 + 1 \cdot (-1) \\ -0.5 \cdot 1 + 0.5 \cdot 0 & -0.5 \cdot 0 + 0.5 \cdot (-1) \\ 0.5 \cdot 1 + (-0.5) \cdot 0 & 0.5 \cdot 0 + (-0.5) \cdot (-1) \\ 1 \cdot 1 + (-1) \cdot 0 & 1 \cdot 0 + (-1) \cdot (-1) \\ 1 \cdot 1 + (-1) \cdot 0 & 1 \cdot 0 + (-1) \cdot (-1) \end{bmatrix} \qquad = \begin{bmatrix} -1 \cdot 0 + 1 \cdot (-1) & -1 \cdot 1 + 1 \cdot 0 \\ -1 \cdot 0 + 1 \cdot (-1) & -1 \cdot 1 + 1 \cdot 0 \\ -0.5 \cdot 0 + 0.5 \cdot (-1) & -0.5 \cdot 1 + 0.5 \cdot 0 \\ 0.5 \cdot 0 + (-0.5) \cdot (-1) & 0.5 \cdot 1 + (-0.5) \cdot 0 \\ 1 \cdot 0 + (-1) \cdot (-1) & 1 \cdot 1 + (-1) \cdot 0 \\ 1 \cdot 0 + (-1) \cdot (-1) & 1 \cdot 1 + (-1) \cdot 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 \\ -1 & -1 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = X_2. \qquad = \begin{bmatrix} -1 & -1 \\ -1 & -1 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = X_2.$$

Therefore, the following equations hold:

$$X_1 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = X_2, \quad X_1 \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = X_2.$$

The first equation corresponds to a reflection, and the second one corresponds to a rotation by 90°. Therefore, we can conclude that the two embedding results are related by orthogonal transformations. These transformations preserve the relative distances and geometry in embedding spaces. Thus, although the two embeddings differ numerically, they represent the same, up to an orthogonal transformation, and equally valid embedding of the graph structure.

3 Question 3

With $A_1 = PAP^T$ and $X_1 = PX$, we have:

$$GNN(PAP^T, PX) = f(\hat{A}_1 X_1 W).$$

We need to find and simplify $\hat{A}_1 = \tilde{D}_1^{-\frac{1}{2}}(A_1 + I)\tilde{D}_1^{-\frac{1}{2}}$, where $\tilde{D}_1 = \sum_j [A_1 + I]_{ij}$ is a diagonal degree matrix. First, using $PP^T = PIP^T = I$, we can compute:

$$A_1 + I = PAP^T + I = PAP^T + PIP^T = P(A + I)P^T = P\tilde{A}P^T.$$

From the previous equation, we now know that $\tilde{D}_1 = \sum_j [P\tilde{A}P^T]_{ij}$. Because $P\tilde{A}P^T$ is simply $\tilde{A}$ with permuted rows and columns, its diagonal degree matrix is permuted in the same way, hence $\tilde{D}_1 = P\sum_j \tilde{A}_{ij}P^T = P\tilde{D}P^T$. Moreover, taking the previous equation to power of $-\frac{1}{2}$ only affects the diagonal matrices and not the permutations, so $\tilde{D}_1^{-\frac{1}{2}} = P\tilde{D}^{-\frac{1}{2}}P^T$.
We can now simplify $\hat{A}_1$ as:

$$\begin{aligned}
\hat{A}_1 &= \tilde{D}_1^{-\frac{1}{2}}(A_1 + I)\tilde{D}_1^{-\frac{1}{2}} \\
&= \left(P\tilde{D}^{-\frac{1}{2}}P^\top\right)\left(P\tilde{A}P^\top\right)\left(P\tilde{D}^{-\frac{1}{2}}P^\top\right) \\
&= P\tilde{D}^{-\frac{1}{2}}I\tilde{A}I\tilde{D}^{-\frac{1}{2}}P^\top \\
&= P\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}P^\top,
\end{aligned}$$

but $\hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}$, so:

$$\hat{A}_1 = P\hat{A}P^\top.$$

We can substitute everything into very first equation and obtain:

$$\begin{aligned}
GNN(PAP^\top, PX) &= f(\hat{A}_1 X_1 W) \\
&= f\big((P\hat{A}P^\top)(PX)W\big) \\
&= f\big(P\hat{A}IXW\big) \\
&= f\big(P\hat{A}XW\big).
\end{aligned}$$

Since P does the permutation of rows, we know that, for some matrix Y:

$$f(PY) = Pf(Y),$$

Now, if we take $Y = \hat{A}XW$, we finally have

$$GNN(PAP^\top, PX) = f(P\hat{A}XW) = Pf(\hat{A}XW) = P \cdot GNN(A, X),$$

which proves permutation equivariance of the GNN layer.

4 Question 4

1. We need to show that $\hat{A}u = \lambda u$, with $\lambda = 1, u_i = \sqrt{\tilde{d}_i}$. We first observe the following:

$$\hat{A}u = \tilde{D}^{-\frac{1}{2}}\,\tilde{A}\,\tilde{D}^{-\frac{1}{2}}u.$$

If we look at i-th entry of vector $\tilde{D}^{-\frac{1}{2}}u$, we have:

$$(\tilde{D}^{-\frac{1}{2}}u)_i = \tilde{d}_i^{-\frac{1}{2}} \cdot \sqrt{\tilde{d}_i} = 1,$$

so $\tilde{D}^{-\frac{1}{2}}u = \mathbb{1}$, i.e., all-ones vector. Consequently, we have:

$$\hat{A}u = \tilde{D}^{-\frac{1}{2}}\,\tilde{A}\,\mathbb{1}.$$

If we now look at i-th entry of vector $\tilde{A}\mathbb{1}$, we have

$$(\tilde{A}\mathbb{1})_i = \sum_{j=1}^{n} \tilde{A}_{ij} \cdot 1 = \sum_{j=1}^{n} \tilde{A}_{ij} = \tilde{d}_i.$$

so $\tilde{A}\mathbb{1} = \tilde{D}\mathbb{1}$, i.e., a vector of node degrees. Again, with power simplification, we have:

$$\hat{A}u = \tilde{D}^{-\frac{1}{2}}\tilde{D}\mathbb{1} = \tilde{D}^{\frac{1}{2}}\mathbb{1}.$$

Now, i-th entry of vector $\tilde{D}^{\frac{1}{2}}\mathbb{1}$ is:

$$(\tilde{D}^{\frac{1}{2}}\mathbb{1})_i = \tilde{d}_i^{\frac{1}{2}} = \sqrt{\tilde{d}_i} = u_i,$$

so $\tilde{D}^{\frac{1}{2}}\mathbb{1} = u$, and $\hat{A}u = u \iff \hat{A}u = \lambda u, \lambda = 1$. Therefore, u is an eigenvector of $\hat{A}$ that corresponds to the eigenvalue $\lambda = 1$.

2. We use the spectral decomposition of $\hat{A}$ to obtain the following:

$$\hat{A} = Q\Lambda Q^\top,$$

where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and Q is a matrix with eigenvectors $q_1, \ldots, q_n$ as its columns. We want to raise both sides to the k-th power, so we first observe how $\hat{A}$ behaves in that case:

$$\begin{aligned}\hat{A}^2 &= \hat{A}\hat{A} \\ &= (Q\Lambda Q^\top)(Q\Lambda Q^\top) \\ &= Q\Lambda I \Lambda Q^\top \\ &= Q\Lambda^2 Q^\top.\end{aligned}$$

We see that Q and Q^T cancel each other (because Q is orthogonal) and that the power only affects Λ. Therefore, raising both sides to the k-th power results in:

$$\begin{aligned}\hat{A}^k &= Q\Lambda^k Q^\top, \\ \Lambda^k &= \mathrm{diag}(\lambda_1^k, \ldots, \lambda_n^k).\end{aligned}$$

We now look at $\Lambda^k, k \to \infty$. From task's description and since the first eigenvalue is the largest one, we have:

$$\lim_{k\to\infty} \lambda_i^k = \begin{cases} 1, & i = 1, \\ 0, & i \geq 2, \end{cases} \qquad \Longrightarrow \qquad \lim_{k\to\infty} \Lambda^k = \mathrm{diag}(1, 0, \ldots, 0).$$

Therefore,

$$\lim_{k\to\infty} \hat{A}^k = \lim_{k\to\infty} Q\Lambda^k Q^\top = Q\,\mathrm{diag}(1, 0, \ldots, 0)\,Q^\top = q_1 q_1^\top,$$

where q_1 is a normalized eigenvector associated to $\lambda_1 = 1$. From Question 4.1, this vector is known and is given by:

$$q_1 = \frac{u}{\|u\|}, u_i = \sqrt{\tilde{d}_i}.$$

Substituting this back, we have:

$$\lim_{k\to\infty} \hat{A}^k = \frac{u}{\|u\|}\frac{u^\top}{\|u\|} = \frac{uu^\top}{\|u\|^2}.$$

Finally, the node representations with $k \to \infty$ are computed as follows:

$$\lim_{k\to\infty} Z^{(k)} = \lim_{k\to\infty} (\hat{A}^k XW) = \left(\frac{uu^\top}{\|u\|^2}\right) XW.$$

For the next question, we also want to compute the node representations for a specific node i (which corresponds to row i):

$$\lim_{k\to\infty} Z_i^{(k)} = \left(\frac{u_i u^\top}{\|u\|^2}\right) XW = \frac{u_i}{\|u\|^2}\left(u^\top XW\right) = \frac{\sqrt{\tilde{d}_i}}{\|u\|^2}\left(u^\top XW\right)$$

3. All node representations converge to the same vector $\frac{(u^\top XW)}{\|u\|^2}$ scaled by degree factor $\sqrt{\tilde{d}_i}$, and therefore, regardless of their initial features, the nodes with the same degree have the same representations.

References

[1] M. Maruf and Anuj Karpatne. Maximizing cohesion and separation in graph representation learning: A distance-aware negative sampling approach. *arXiv preprint arXiv:2007.01423v1*, 2020.